

A GPU-Accelerated Hydrodynamics Solver For Atmosphere-Fire Interactions

Jhamieka Greenwood

Florida State University
Department of Scientific Computing
Tallahassee, Florida, USA
jsg15h@my.fsu.edu

Dr. Bryan Quaife

Florida State University
Department of Scientific Computing
Tallahassee, Florida, USA
bquaife@fsu.edu

Dr. Kevin Speer

Florida State University
Geophysical Fluid Dynamics Institute
Tallahassee, Florida, USA
kspeer@fsu.edu

KEYWORDS

augmented reality, computational fluid dynamics, fire dynamics, GPU, Navier Stokes, prescribed fire

ACM Reference Format:

Jhamieka Greenwood, Dr. Bryan Quaife, and Dr. Kevin Speer. 2022. A GPU-Accelerated Hydrodynamics Solver For Atmosphere-Fire Interactions. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '22 Posters)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532719.3543263>

1 INTRODUCTION

Wildfires are one of the deadliest and costliest natural disasters in the United States and around the globe. Every year the US experiences progressively more dangerous, destructive, and unpredictable wildfires. The 2021 Bootleg Fire swept through southern Oregon between July 6th and August 15th. Its fastest spread rate was 1,000 acres per hour, burning more than 400,000 acres within two weeks. It is clear that there is a precedented need for more ways to increase predictability and mitigate fire risks. In the Southeast US, government agencies and private landowners use prescribed fire as a land management tool. Prescribed fire reduces the risk of wildfires, destroys invasive vegetation, and promotes biodiversity. Understanding processes associated with prescribed fire and wildfire spread is the goal of our research.

A fundamental process to understand fire spread is the atmospheric flow. Building computational tools to simulate this complex flow has several challenges, including boundary layer effects, resolving vegetation and the forest canopy, conserving fluid mass, and describing fire-induced flows. To understand large scale dynamics, we first must understand what happens at smaller scales. By studying scaled-down models for fire dynamics, we investigate the fundamental processes critical to fire spread. One of the most important factors is the complex coupling of the atmospheric flow and the combusting environment. In particular, what variables influence fire-induced flows, and how do they contribute to fire spread? Moreover, we develop a framework that allows for real-time updates to the combusting environment. This last step is critical so that the results can be interpreted and used by fire managers.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '22 Posters, August 07-11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9361-4/22/08.

<https://doi.org/10.1145/3532719.3543263>

2 METHODS

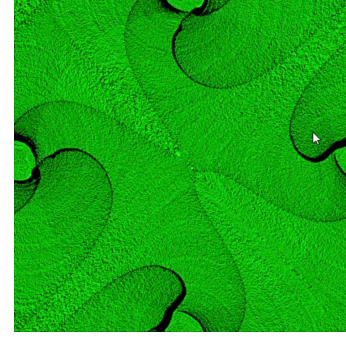


Figure 1: Particles (green) being transported by a solution of Equation (1). There are two delta functions with centers near the middle of the domain.

We propose a model based on a modified version of the two-dimensional Navier-Stokes equations:

$$\nabla \cdot \mathbf{u} = - \sum_{i=1}^N \delta(\mathbf{x} - \mathbf{x}_i) \quad (1a)$$

$$\frac{\partial \mathbf{u}}{\partial t} = -(\mathbf{u} \cdot \nabla) \mathbf{u} - \frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \frac{1}{\rho} \mathbf{f}, \quad (1b)$$

where $\mathbf{u}$ is the fluid velocity, p is the pressure, $\mathbf{f}$ is the external forces, and $\mathbf{x}_i$ are sites of combustion fuels. Using the Dirac mass, δ , is a novel approach to convert the two-dimensional horizontal momentum into vertical momentum through the buoyant plume at the locations $\mathbf{x}_i$, $i = 1, \dots, N$. As an extension of the GPU-accelerated hydrodynamic solver of Stam [Stam 1999], we solve equation (1) using operator splitting and pressure projection (see Figure 1). This solver requires methods to solve an advection equation, a diffusion equation, and a projection step [Chorin 1968]. The solvers for the advection and diffusion equations are discussed in the next section. The projection step maps a velocity field $\mathbf{w}$ to the closest velocity field to $\mathbf{w}$ that satisfies the mass equation in equation (1a). Following the classical projection operator, the modified projection operator is $P\mathbf{w} = \mathbf{w} - \nabla q$, where

$$\nabla^2 q = \nabla \cdot \mathbf{w} + \sum_{i=1}^N \delta(\mathbf{x} - \mathbf{x}_i). \quad (2)$$

Equation (2) is solved by writing q as the sum of the fundamental solution of the Laplacian, ∇^2 , centered at $\mathbf{x}_i$, and a regular term that satisfies Laplace's equation.

3 IMPLEMENTATION

Many software packages used in fire simulations are computationally expensive and take hours to run to completion. It is critical that algorithms used in fire dynamics be faster in real-time to be transitioned into an operational tool. Our GPU-accelerated hydrodynamics solver is based on modified algorithms of Stam's Stable Fluids system [Stam 1999]. The solver uses CUDA kernels to efficiently apply the following routines on a uniform grid at each time step: introduce a user-specified force, advect the velocity, diffuse the velocity, and project the results of the velocity field that satisfies our modified mass equation.

Solutions of the advection equation, the diffusion equation, and the projection equation (equation (2)) are solved with high-order accuracy. Similar to Stam [Stam 1999], we assume the solution of (1) is doubly-periodic. The advection equation is solved with a semi-Lagrangian method that uses a local bilinear interpolant. The diffusion and projection equations are solved using Fourier

methods. Finally, a smooth cutoff function guarantees that the solution remains doubly-periodic.

4 CONCLUSION AND FUTURE WORK

We utilize the interoperability of CUDA and OpenGL to display our grid domain through a particle-based system and to incorporate interactivity. A point-source convective fire is simulated through user input at any point of the solver domain, and we can investigate the atmosphere-fire interactions around this point. Future work will couple the solver with a topographical map on a 5 x 4-foot sand table filled with reflective sand. Our goal is to examine the elevation effects of point-source fires spreading across a terrain and adding various parameters and conditions.

REFERENCES

- Alexandre Joel Chorin. 1968. Numerical solution of the Navier-Stokes equations. *Mathematics of computation* 22, 104 (1968), 745–762.
- Jos Stam. 1999. Stable Fluids. In *Proceedings of the 26th Annual Conference on Computer Graphics and Interactive Techniques*. 121–128.